Husin Bin Hamdan

AUTOMATION AND ROBOTIC STUDENT

+60 19-377 8534 @ 020husin.hamdan@gmail.com

Lot 1385 Kanchong Darat, Banting 42700 Selangor

EXPERIENCE

Automation Robotic Internship Trainee

TXMR SDN BHD

06/2023 Malaysia

- Contribute to the installation and commissioning of a robotic arm welding system, reducing setup time by 15% through system calibration, wiring, and on-site testing.
- improved system reliability by troubleshooting and maintaining sensors and actuators, reducing unexpected downtime during testing phases.
- Executed end-to-end electrical wiring between control panel and Robotic Arm, ensuring 100% signal integrity and safety during commissioning tests.

Mechanical Design & Procurement Assistant (Part-Time)

FICES Technology

9/2025 - 04/2026 Malaysia

- Designed and developed prototypes for SMART Agriculture & IoT Monitoring Database system (TerraSens).
- Selected and procured technical component and materials, ensuring compatibility with design specifications.
- Implemented product improvements to enhance functionality, manufacturability, and overall performance.
- Collaborated with team members to iterate designs and optimize production processes.

Treasurer Robotique Society Club

UniKL Malaysia France Institute

03/2024 - 04/2026 Malaysia

University institute focusing on engineering technology

- Managed and monitored the club’s financial records, including tracking expenses for club activities and events
- Maintained organized documentation of financial transactions to ensure transparency and accountability
- Prepared financial records and reports to support budget claims and funding requests for the club

Head of Mechanical Team - Robotics Competition

UniKL Malaysia France Institute

03/2024 - 04/2026 Malaysia

University institute focusing on engineering technology

- Led mechanical design and fabrication of competition robot, improving movement stability and achieving successful task execution during ABU Robocon Malaysia 2025 trials.
- Responsible for mechanical design, prototyping, testing, and tuning of the robot’s mechanisms.
- Conducted R&D and collaborated with the team to optimize the robot’s mechanical performance for competition

SUMMARY

Bachelor of Engineering Technology (Automation and Robotics) student with hands-on experience in robotic systems, sensor integration, and embedded control. Experienced in industrial automation environments, including robotic arm commissioning, electrical wiring, and system troubleshooting. Strong interest in IoT systems, real-time monitoring and manufacturing, and manufacturing optimization, with proven ability to lead technical teams and deliver functional engineering solutions.

TECHNICAL SKILLS

Mechanical & Design

- Solidworks / Catia / Inventor
- CAD modelling & prototyping

Electrical

- Circuit wiring, Troubleshooting
- Electrical Wiring and Control Panel Integration

Embedded Systems & Microcontroller Applications

- Python, C/C++,
- Microcontroller-based System Design and Development.
- Real-Time data Acquisition and Embedded Control

Automation & Control Systems

- Robotic system integration and commissioning
- Troubleshooting and System diagnostics
- PLC basic (if applicable)
- IOT integration (Node-RED)
- Sensor integration & Pneumatic Application

Data & Analytics

- Microsoft Excel (Pivot Table, Dashboard)
- Power BI (Basic visualization)

PROJECTS

Mecanum Wheel Mobile Robot (Slope Navigation)

- Designed and develop a Mobile Robot capable of navigating 5 - 12 degree inclined surfaces
- Integrated IMU sensor for slope detection
- Reduced Wheel slippage by implementing IMU-based feedback control and motor load monitoring.

Combat Robot Development (8Kg Class) Robattle Competition (UniKL MFI)

- Performed structural simulation and impact analysis to evaluate stress distribution and improve fram durability.
- Executed fabrication and assembly of component, ensuring dimensional accuracy and structural reliability

EDUCATION

Bachelor of Engineering Technology (Automation and Robotics) with Honours

UniKL Malaysia France Institute

03/2024 - 03/2027 Malaysia

Diploma of Manufacturing Engineering (Automation and Robotics)

Kolej Kemahiran Tinggi MARA Kuantan

09/2020 - 09/2023 Malaysia